Bandwidth Measurement and Analysis

SDN Mini Project Report

Name: Anusha Gupta **SRN:** PES1UG24CS073
Subject: Software Defined Networking
Tool Used: Mininet + iPerf

1. Introduction

Software Defined Networking (SDN) is a modern networking approach that separates the control plane from the data plane, allowing network administrators to manage network behavior programmatically through a centralized controller. Unlike traditional networks where each switch makes its own routing decisions, SDN enables dynamic, policy-driven network management.

This project explores how different network topologies affect bandwidth performance within an SDN environment. Using Mininet — a widely used network emulator — three distinct virtual network topologies were built and tested: Single, Linear, and Tree. The iPerf tool was then used to measure TCP throughput between various host pairs in each topology.

The primary goal of this experiment is to understand how the structural complexity of a network (number of switches, hops, and shared links) impacts the actual data transfer rate between hosts, which is a critical consideration in the design of real-world networks.

2. Objectives

The key objectives of this mini project are as follows:

- Set up and run different network topologies (Single, Linear, Tree) using Mininet.
- Verify host connectivity using the pingall command in each topology.
- Measure TCP bandwidth between host pairs using the iPerf network performance tool.
- Compare throughput results across topologies and analyze the effect of network complexity on bandwidth.
- Draw meaningful conclusions to guide informed network design decisions.

3. Tools and Environment

3.1 Mininet

Mininet is an open-source network emulator that creates a realistic virtual network — including hosts, switches, controllers, and links — on a single Linux machine. It uses lightweight virtualization (process-based) to simulate end hosts and Open vSwitch (OVS) for software-defined switching. Mininet allows rapid prototyping of SDN topologies without requiring physical hardware.

3.2 iPerf

iPerf is a standard network testing tool used to measure maximum achievable bandwidth on IP networks. It works in a client-server model: a host acts as the server (listening mode) while another host acts as the client and sends data. The tool reports transfer rate (throughput) in Gbps, providing a clear picture of link utilization capacity.

3.3 Open vSwitch (OVS)

Since no external OpenFlow controller was configured in this experiment, Mininet automatically fell back to Open vSwitch (OVS) bridge mode. OVS is a production-grade, multilayer virtual switch that supports standard management interfaces and allows fine-grained programmatic control of network traffic.

4. Methodology

The experiment was conducted in the following sequence for each topology:

1. Mininet was launched with the desired topology using the sudo mn command with appropriate parameters.
1. The pingall command was executed to confirm that all hosts could communicate with each other, validating correct network setup.
1. iPerf was started in server mode on a destination host using the command: iperf -s.
1. iPerf was run in client mode on the source host pointing to the destination IP using: iperf -c <destination_IP> -t 5.
1. The reported throughput (in Gbps) was recorded for each source-destination pair.
1. The above steps were repeated for all topology types and multiple host pairs.

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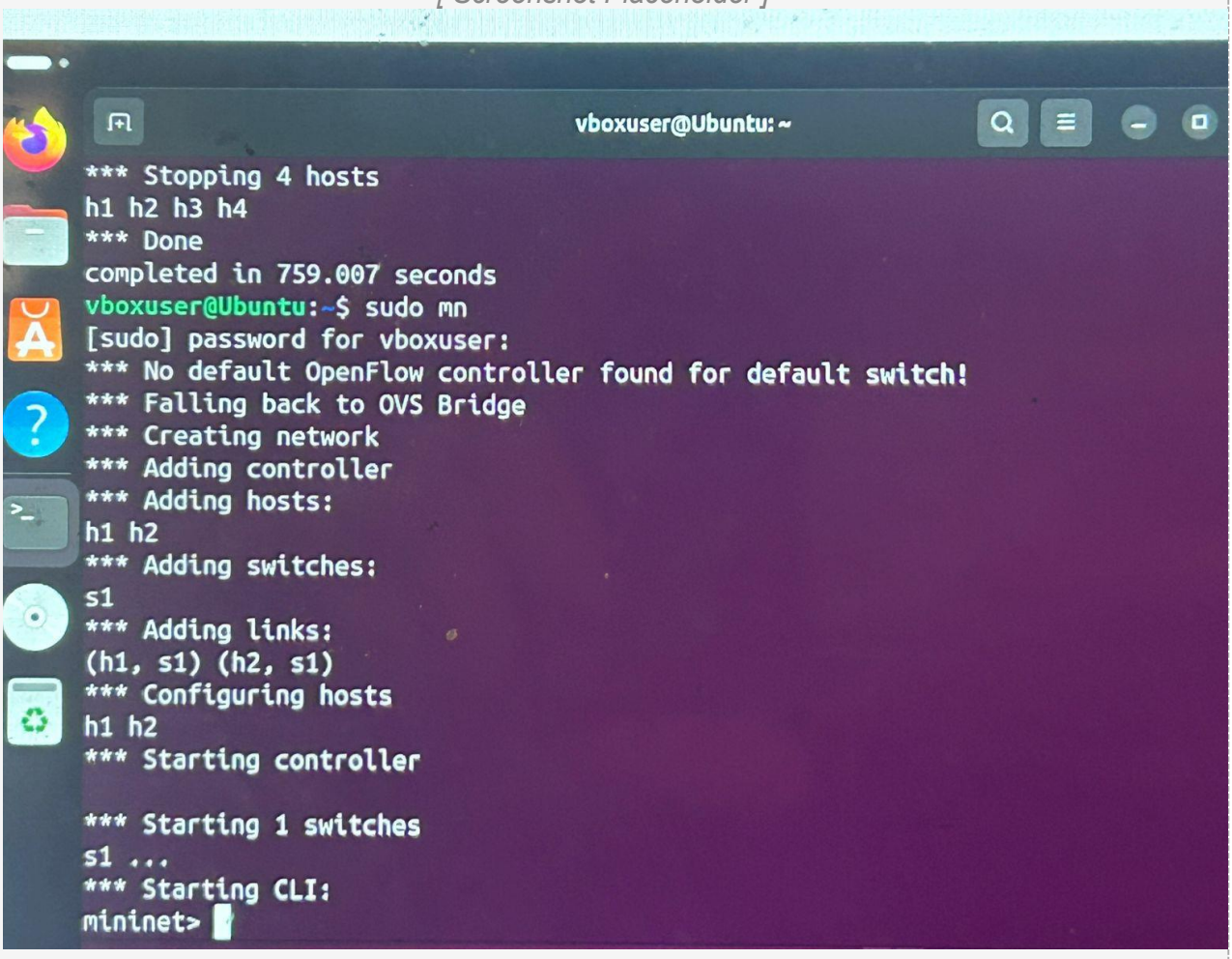

Figure 4: Mininet Running — Terminal Screenshot

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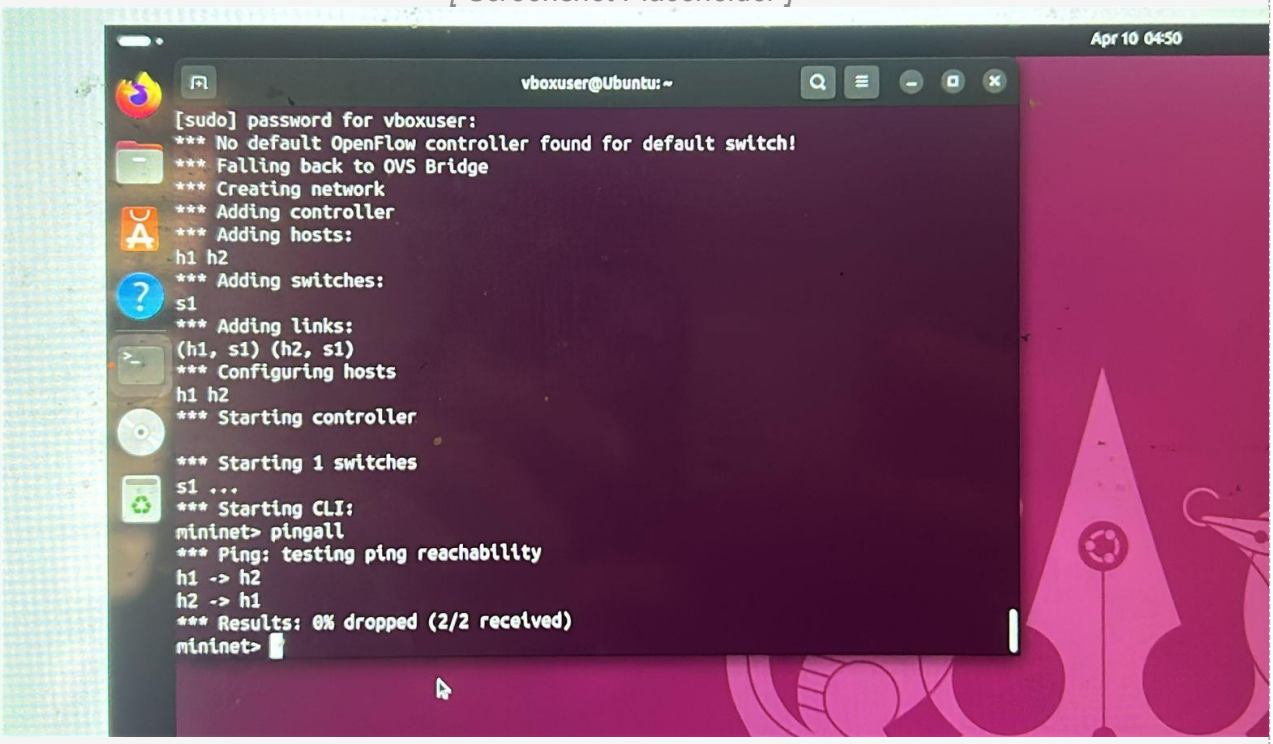

Figure 5: Pingall Command Output Showing 0% Packet Drop

5. Results

The table below presents the iPerf throughput results recorded for each topology and host pair combination. Throughput is measured in Gigabits per second (Gbps).

Topology	Source	Destination	Throughput (Gbps)
Linear	h3	h1	18.3
Linear	h3	h2	19.9
Linear	h1	h2	18.9
Single	h3	h1	21.2
Single	h3	h2	20.6
Single	h1	h2	18.4
Tree	h4	h1	18.1

Topology	Source	Destination	Throughput (Gbps)
Tree	h2	h1	21.8
Tree	h2	h4	17.7
Tree	h1	h4	10.3

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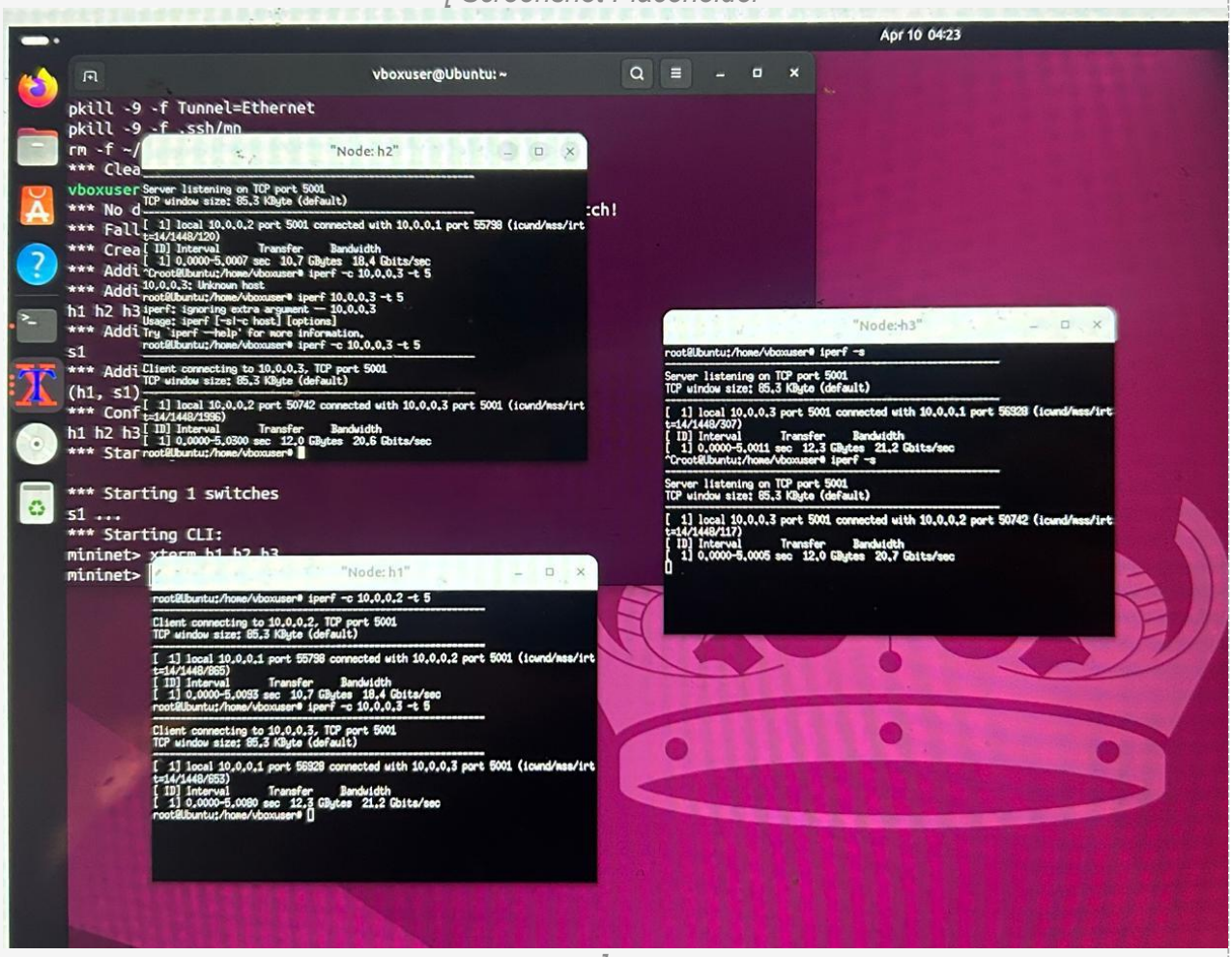

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Figure 6: iPerf Output — Single Topology

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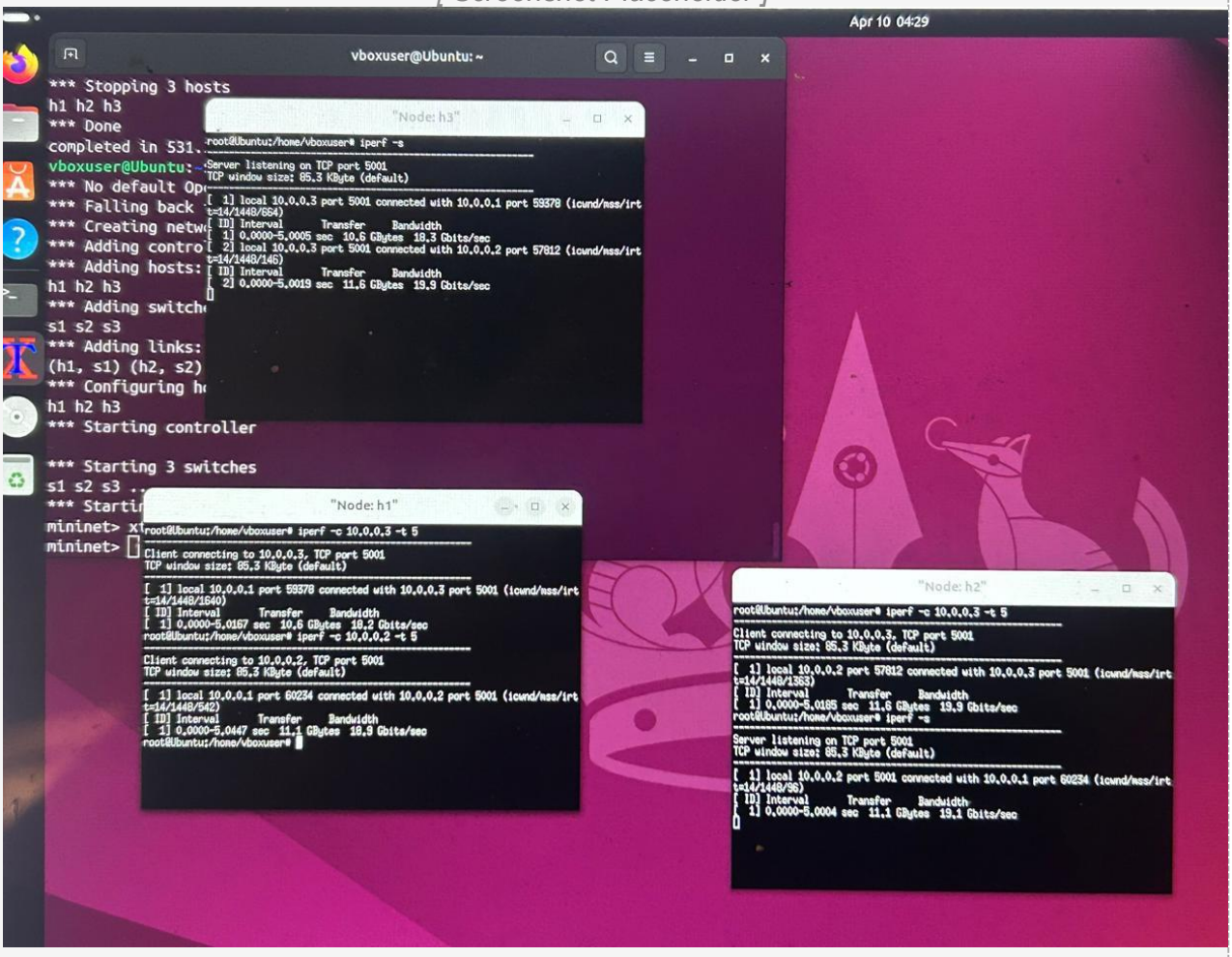

Figure 7: iPerf Output — Linear Topology

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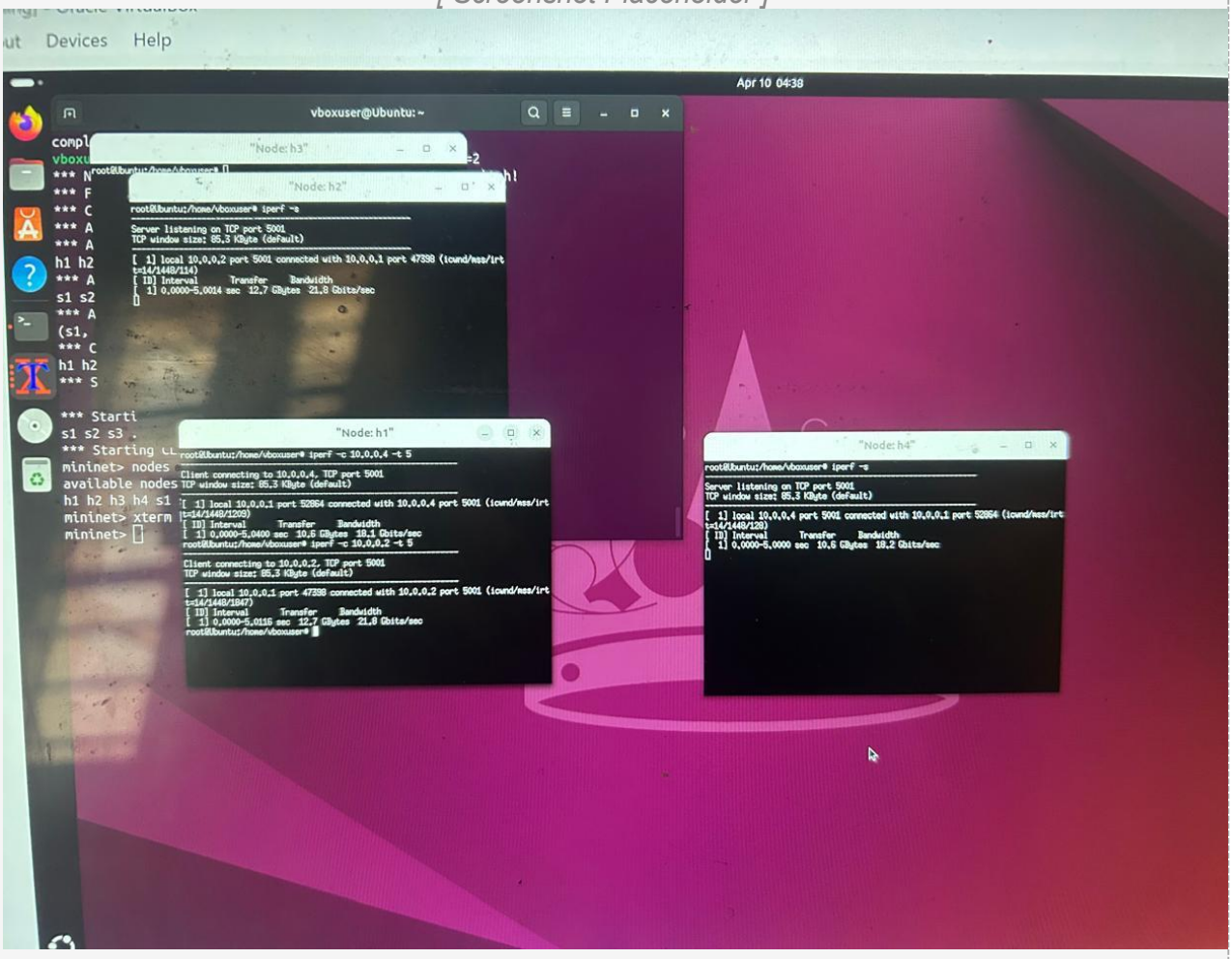

Figure 8: iPerf Output — Tree Topology

6. Analysis and Discussion

7.1 Single Topology

The Single switch topology consistently recorded the highest throughput values — reaching up to 21.2 Gbps (h3 → h1) and 20.6 Gbps (h3 → h2). Since all hosts are connected to a single switch, there is only one hop for any host-to-host communication. This minimizes latency and eliminates congestion from shared intermediate links, allowing the full bandwidth of the virtual links to be utilized.

7.2 Linear Topology

The Linear topology showed slightly lower throughput — ranging from 18.3 to 19.9 Gbps. Although still high in absolute terms, the sequential chain of switches means that data must pass through multiple forwarding stages. Each additional hop introduces a small processing and forwarding delay. Shared intermediate links can also become bottlenecks if multiple flows compete for the same path.

7.3 Tree Topology

The Tree topology exhibited the most variable performance. Host pairs connected within the same subtree (e.g., h2 → h1 at 21.8 Gbps) showed high throughput comparable to the Single topology. However, pairs that required traversal through the root switch — such as h1 → h4, which recorded only 10.3 Gbps — suffered a significant bandwidth reduction. This degradation is attributed to the increased number of hops and the shared uplink capacity at higher levels of the tree hierarchy.

7.4 General Observation

Across all three topologies, a clear inverse relationship is observed between network complexity and throughput. As the number of intermediate switches and hops increases, effective bandwidth decreases. Topology design is therefore a critical factor in SDN performance optimization — simpler topologies or those that minimize path length between frequently communicating hosts will deliver better bandwidth utilization.

7. Conclusion

This project successfully demonstrated the impact of network topology on bandwidth performance in a Software Defined Networking environment using Mininet and iPerf.

The Single switch topology delivered the best average throughput due to its simplicity and absence of intermediate hops. The Linear topology showed a moderate performance reduction caused by multi-switch forwarding paths. The Tree topology produced the most variable results — near-optimal performance for hosts within the same branch, but significantly lower throughput for cross-branch communication.

These results underscore the importance of thoughtful network design in SDN environments. When performance and low latency are priorities, minimizing the number of hops between critical hosts — either through topology choice or intelligent routing policies — is essential. SDN's programmability provides a powerful advantage here, enabling dynamic path optimization that static networks cannot achieve.

Future work could explore the impact of bandwidth throttling, custom OpenFlow controllers, and traffic shaping policies on these topologies to further characterize SDN performance.